

COMPUTER ARCHITECTURE

ASSIGNMENT – 1

Instructions:

- Attempt all questions.
 - Show complete steps for numerical problems.
 - Draw neat and labelled diagrams wherever required.
 - Use suitable examples wherever asked.
1. Define Computer Architecture and Computer Organization. Explain the difference between them.
 2. What is a register? Explain why registers are required in a CPU. What is Register Transfer Language (RTL)? Explain its notation with suitable examples.
 3. Explain the concept of a common bus system.
 4. Explain a common bus system using multiplexers with a neat diagram.
 5. Explain the stored-program organization with a neat diagram.
 6. Given $R1 = 1011$ and $R2 = 0101$, perform $R3 \leftarrow R1 + R2$. Find $R3$.
 7. Given $AC = 1010$ and $DR = 0011$, perform $AC \leftarrow AC + DR$. Find the new contents of AC .
 8. Given $R1 = 1101$, perform $R2 \leftarrow R1$ and then $R3 \leftarrow R2$. Find the final values of $R1$, $R2$ and $R3$.
 9. Given $AC = 1100$ and $DR = 0011$, perform $AC \leftarrow AC - DR$ using 2's complement.
 10. Given $R1 = 1111$, perform $R1 \leftarrow R1 + 1$ using 4-bit arithmetic. State the result and carry.
 11. Four registers contain $R0 = 1010$, $R1 = 0111$, $R2 = 1100$ and $R3 = 0011$. If the common bus selects $R2$ and $R0$ is the destination, determine the bus value and new $R0$.
 12. Perform the following sequence for $AC = 1010$: $AC \leftarrow AC + 0011$; $AC \leftarrow AC - 0010$. Find final AC .
 13. Given $PC = 1111$, perform $PC \leftarrow PC + 1$ using a 4-bit register. Explain the result.
 14. Explain a 4-bit binary parallel adder with a neat block diagram.
 15. Explain the design and working of a 4-bit arithmetic circuit.
 16. What is an Arithmetic Logic Unit (ALU)? Explain its major functions.
 17. Explain logical shift operations: logical left shift and logical right shift.
 18. Perform the binary addition: $101101 + 110011$.
 19. Perform binary subtraction: $101101 - 011011$.
 20. Represent -37 using an 8-bit 2's complement representation.
 21. Convert these 8-bit 2's complement numbers into decimal: 11100101 , 10011011 and 11010110 .

22. Given $A = 10110110$ and $B = 11001101$, calculate $A \text{ AND } B$, $A \text{ OR } B$, $A \text{ XOR } B$ and $\text{NOT } A$.
23. Given $A = 11001100$ and $B = 10101010$, calculate $A \text{ AND } B$, $A \text{ OR } B$ and $A \text{ XOR } B$.
24. Given $R = 11010110$, perform $R \leftarrow \text{shl } R$ and then $R \leftarrow \text{shr } R$. Find the final contents.
25. Given $R = 10011001$, perform logical left shift, logical right shift and arithmetic right shift.
26. For $A = 1001$ and $B = 0101$, perform $A + B$, $A - B$, $A \text{ AND } B$, $A \text{ OR } B$ and $A \text{ XOR } B$.
27. A 4-bit ALU has $A = 1110$ and $B = 0011$. Calculate $A - B$ using 2's complement.
28. An 8-bit accumulator contains $AC = 10110101$. Perform $AC \leftarrow AC + 00000101$ and then $AC \leftarrow \text{shl } AC$. Find the final value.
29. A 4-bit register initially contains 1011. Perform sequentially: $R \leftarrow R + 0011$; $R \leftarrow \text{shr } R$; $R \leftarrow R \text{ XOR } 1100$. Find the final contents.
30. Given $AC = 1010$ and $DR = 0110$, perform sequentially: $AC \leftarrow AC + DR$; $AC \leftarrow \text{shl } AC$; $AC \leftarrow AC$.
31. A 4-bit accumulator contains 1101. Perform: $AC \leftarrow AC + 0010$; $AC \leftarrow AC \text{ XOR } 1111$; $AC \leftarrow \text{shr } AC$. Find the final contents.
32. Convert decimal numbers +18 and -18 into 8-bit sign-magnitude, 1's complement and 2's complement representations.
33. Perform the following sequence on $R = 10110010$: logical right shift, circular left shift, and arithmetic right shift. Show the result after each operation.
34. Explain with a real-life example how the ALU, registers and common bus work together when a calculator performs $25 + 15$.
35. Explain with a real-life example where shift operations are used in computers or digital devices.
36. Explain how a common bus system reduces the number of direct connections between registers.
37. Explain why an accumulator is useful when performing a sequence of arithmetic operations.